ARTHUR ARIES

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Computer Vision & Embedded AI Engineer

French AI Engineer with over 6 years of experience specializing in Computer Vision and Embedded Deep Learning. Adept at bridging the gap between advanced ML research and robust software engineering.

- **Edge AI Expertise:** Proven track record in optimizing complex models (TensorRT, ROS 2) for real-time applications on highly resource-constrained hardware (< 50ms latency).
- **End-to-End MLOps:** Strong capability to design standalone architectures and internal tooling (AimFlow, AimLab) closing the loop from active learning to deployment.

Seeking new engineering challenges as a Computer Vision / Embedded AI Engineer (open to broader AI/ML engineering challenges), with a primary focus on opportunities in Alsace or full-remote roles, while remaining open to positions across France and international opportunities.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, TypeScript.
- **AI & Computer Vision:** PyTorch, TensorFlow, OpenCV, SMP, SAM / SAM 2, DeepLabV3+, YOLO.
- **Embedded & Deployment:** NVIDIA Jetson (Orin NX), NXP i.MX 8, TensorRT, ONNX, ROS 2 (Isaac ROS), GStreamer.
- **Tools, Web & MLOps:** Linux, Git, Docker, MLflow, Optuna, FastAPI, Tauri, React, SQLite.

WORK EXPERIENCE

Consultant – AI & Computer Vision Engineer — Nov. 2020 – Present
Technology & Strategy | Strasbourg (FR) & Stuttgart (DE)

Client Project: Noremat (Autonomous Roadside Mowing) — 2026

- Designed and engineered an embedded semantic segmentation solution (7 classes, DeepLabV3+, EfficientNet) deployed on NVIDIA Jetson Orin NX for lateral autonomous guidance.
- Optimized and deployed the pipeline under ROS 2 (TensorRT FP16, Isaac ROS), achieving a stable 10 FPS (< 100 ms end-to-end latency) and a ∼75% mIoU on test benchmarks.
- Curated a 3,000+ image proprietary dataset leveraging automated zero-shot labeling (SAM), reducing critical road/shoulder class confusion by over 40%.

Client Project: ARQUUS Defense (Path Detection) — 2024 – 2025

- Developed a semantic segmentation demonstrator (DeepLabV3, MobileNetV2) for autonomous off-road military vehicle path detection.
- Rewrote the complete training, benchmarking, and evaluation pipeline from scratch using PyTorch and SMP across 15+ architectures, securing an inference time under 30 ms and mIoU exceeding 0.70.
- Cleaned and structured a large-scale heterogeneous dataset (5,000+ images) via automated and manual double-filtering techniques to eliminate environmental biases.

R&D Project: AimFlow (MLOps Platform) — 2025 – Present

- **Sole Engineer:** Conceived and developed a standalone desktop MLOps platform from scratch (Tauri, React, PyTorch) to fully automate the model training, evaluation, and deployment lifecycle.
- Integrated a unified model factory supporting 15+ segmentation and detection architectures, featuring MLflow tracking, hyperparameter tuning (Optuna), and ONNX/Docker export.
- Built a high-performance desktop UI for interactive YAML configuration management and real-time video inference testing.

R&D Project: AimLab (Annotation Tooling) 2026 – Present

- **End-to-End Ownership:** Designed and delivered a production-ready standalone desktop application for semantic annotation, deployed internally to accelerate R&D loops.
- Embedded a microservice wrapping SAM 2 (Meta AI) alongside traditional computer vision algorithms (Watershed, smart brush) for zero-shot assisted annotation.
- Implemented an active learning pipeline (FastAPI, React, SQLite) allowing corrected inferences to be directly re-injected into the dataset.

Client Project: Bosch (Real-Time Edge AI) 2020 – 2024

- Deployed real-time object detection models (YOLO, NanoDet) onto highly resource-constrained target hardware (NXP i.MX 8 with NPU, NVIDIA Jetson with GPU).
- Created robust pseudo-labeling algorithms to exploit massive unlabelled agricultural image data, slashing manual annotation overhead by 50% to 80%.
- Conducted extensive profiling and optimization to balance the FPS/Memory/Latency trade-off, securing ultra-low latencies ($<$ 50 ms end-to-end).

Data Scientist Intern Apr. 2020 – Oct. 2020
Alstom | Tarbes, France

- Developed embedded Machine Learning models for KZ8A freight locomotives to predict and mitigate critical wheel slip-stick anomalies.
- Migrated the legacy C/Matlab codebase into a modern Python ecosystem to streamline R&D experimentation and model performance evaluation.
- Designed and trained dense neural networks (TensorFlow, Keras) to forecast anomaly severity within a 20-to-100ms horizon using unsupervised sensor analysis (K-Means, T-SNE).

Web & Android Developer Intern Apr. 2019 – Aug. 2019
TokyoStay | Tokyo, Japan

- Built a responsive web platform and developed a companion Android application for international apartment rentals, enhancing global user acquisition.

PERSONAL PROJECTS

- **Flavoria - 2D Game Development (*Godot, GDScript*):** Designing and programming a monster-collecting dungeon crawler from scratch, implementing Object-Oriented principles, state machines, and custom entity behaviors.
- **Top 14 Fantasy Rugby Optimizer (*Python*):** Developed an algorithmic optimization tool for "La Grande Mêlée" fantasy rugby, maximizing team composition points under strict budget and positional constraints.

EDUCATION

EISTI (Graduate School of Computer Science) 2016 – 2020
Master's Degree in Computer Science – AI & Deep Learning *Pau, France*

EPF Graduate School of Engineering 2014 – 2016
Integrated Scientific Preparatory Program *Montpellier, France*

CERTIFICATIONS & LANGUAGES

- **Certifications:** IBM Data Science Professional Certificate, IBM Applied AI, UC Irvine Internet of Things (IoT) & Embedded Systems (*Coursera*).
- **Languages:** English (Fluent - C1), French (Native), Chinese (Notions), Spanish (Notions).

INTERESTS & ACTIVITIES

- **Sports & Competition:** Padel, Football, Formula 1, Esports.
- **Game Development & Strategy:** Passionate about complex system design and algorithmic puzzle-solving.
- **Curiosity & Learning:** Avid reader, continuously exploring new trends in AI and modern software architecture.